Using the BL5A GUI

This is the operator's guide to the windows: what each part shows, what each button sends, and the two experiments from start to end. To build the programs and set them up at the beamline, see install.md.

The screenshots were taken against the simulation (`bl5a-sim`), so the ring current, shutter and temperature fields read "—" in them.

Before you start

The GUI is a client. On its host, user systemd keeps spec-ioc, `bl5a-qs` (the queue server, which runs the plans) and `bl5a-integrator` (which integrates each frame) running (install.md, section 5); the beamline runs the mar345 IOC. You start SPEC in server mode, then the GUI with the **BL5A** shortcut on the desktop or in the applications menu. `systemctl --user status 'bl5a*'` shows whether the services run.

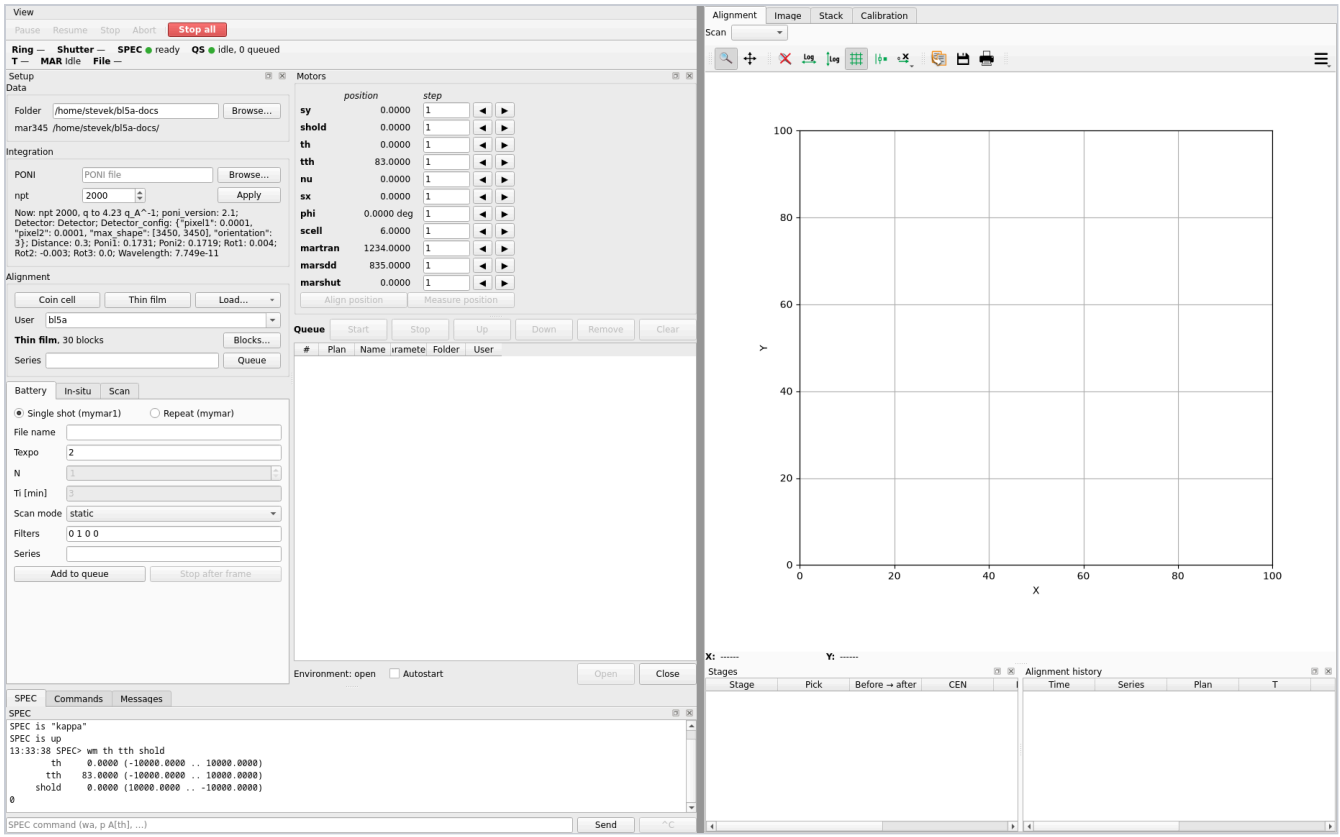
In a terminal, `deploy/bl5a-gui` starts the same GUI and takes these options:

Option	Default	What it is
`--qs-control`	`tcp://localhost:60715`	bl5a-qs control socket
`--qs-documents`	`tcp://localhost:60725`	bl5a-qs document stream
`--integrator-rep`	`tcp://localhost:60735`	integrator requests
`--integrator-pub`	`tcp://localhost:60745`	integrator new rows and errors
`--user`	`bl5a`	the name in the queue's User column

You can close the GUI at any time. The queue keeps running in bl5a-qs and the stacks stay with the integrator. The GUI starts again with an empty Alignment tab and empty Commands and Messages panes: they are filled from what the plans report while the GUI is open. The Stack tab asks the integrator, so it shows every stack it holds.

The windows

The GUI opens two windows side by side: the main window, which runs the experiment, and the plots window.



The main window, left, and the plots window at rest

Area	What is in it
Top	the View menu, the run buttons (Pause, Resume, Stop, Abort, Stop all) and the status line
Left	the Setup pane: the Data and Integration boxes, the Alignment box, the Battery, In-situ and Scan forms
Centre	the Motors pane over the Queue
Bottom	the SPEC console, the Commands the plans sent, the GUI's Messages, as tabs
Plots window	the Alignment tab: the scan plot, with the Stages of the running alignment and the Alignment history below it; the Image, Stack and Calibration tabs

The Stages and Alignment history panes belong to the Alignment tab, so the Image, Stack and Calibration tabs have the whole plots window to themselves.

The first start puts the plots window beside the main window, as tall, out to the right edge of the screen; with a second screen it can take that screen. Closing it only hides it: **View → Plots window** shows it again. Closing the main window closes both.

Every pane has a title bar. Drag the edge between two panes, or between a pane and the queue or the scan plot, to resize them; a pane smaller than its contents scrolls them. Drag a title bar to another side, or onto another pane to share its place as a tab; the Stages and Alignment history panes move only inside the Alignment tab. The button left of × on the title bar, or a double click on the title bar, floats the pane as a window of its own, which can go to another screen; the same again docks it back. × closes the pane. The **View** menu, also on a right click on a docked pane's title bar, shows a closed pane again, and **Reset layout** puts every pane and the plots window back where the first start puts them. The next start opens both windows at the size, place and layout they had when closed, kept in `~/.config/bl5a/bl5a-gui.conf`.

Who drives SPEC

SPEC runs one command at a time. While the queue runs a plan, the plan owns SPEC, so the GUI greys out everything that would send SPEC a command of its own: the motor tweak buttons, their step fields and the SPEC console's entry.

They are live only when all of these hold:

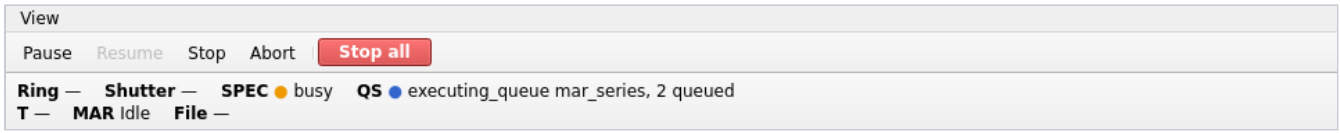
- bl5a-qs is idle and runs nothing;
- spec-ioc reports SPEC connected (`BL5A:SPEC:Connected`);
- SPEC is ready (`BL5A:SPEC:Busy`);
- no console command is running (`BL5A:SPEC:Cmd:Busy`).

The queue itself can be edited at any time.

Status line



The status line at rest



The status line while a mar_series runs

Field	Shows
Ring	the ring current, mA (`BL5A:userCalcOut1.VAL`)
Shutter	the front-end shutter; its PV is not configured yet, so "—"
SPEC	● ready (green), ● busy (orange), ● disconnected (red); "—" when spec-ioc does not answer
QS	the queue server's state, the running plan, how many items wait; "environment closed" and "stopping after the frame" when they apply; ● not answering (red) when bl5a-qs does not reply
T	the temperature as readback → setpoint, and the ramp; "—" until the temperature PVs are configured
MAR	the mar345 detector state (`DetectorState_RBV`)
File	the full path of the last frame; you can select and copy it

The QS dot is green when idle, orange when paused and blue otherwise.

Run buttons

Button	Live when	What it does
Pause	a plan runs	pauses at once; a moving motor is stopped
Resume	paused	goes on
Stop	a plan runs or is paused	ends the plan, recorded as a success

Button	Live when	What it does
Abort	a plan runs or is paused	asks first, then ends the plan, recorded as aborted
Stop all (red)	always	does not ask: sends SPEC ^C, which stops every motor SPEC moves, a count and SPEC's command, and aborts the plan if one runs or is paused

However a plan ends, bl5a-qs then interrupts a SPEC command still running, stops a count still running and closes the mar345 shutter.

Stop all is for when something goes wrong: a motor heading the wrong way, a macro run from SPEC's own terminal, a plan doing what it should not. The ^C reaches SPEC through spec-ioc whatever qs is doing, so it works with qs stopped too. If spec-ioc is not connected, a warning says SPEC got no ^C; stop the motors at SPEC's own terminal then.

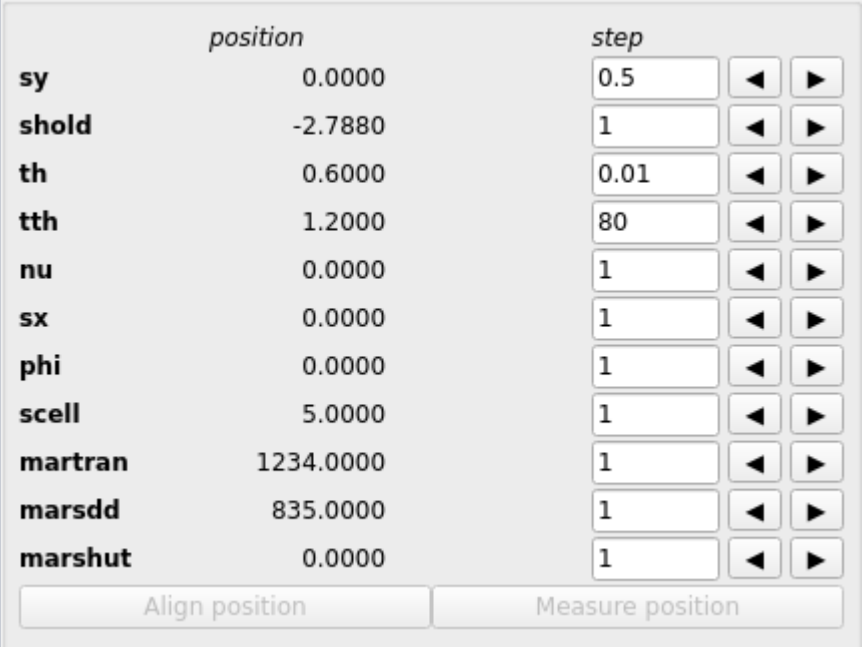
The plans do not replay a SPEC macro that was cut in half. After **Pause**, **Resume** fails the plan with "plan interrupted: a paused plan cannot resume, abort it"; end it with **Stop** or **Abort**. To end a battery series cleanly, use **Stop after frame** on the Battery form instead: that pause lands between frames, where the series can go on.

A motor's stop reaches SPEC as `abort_all`, which stops every motor SPEC is moving.

The queue's own **Stop** is different: it lets the running item finish and then stops the queue.

The mar345's motors (`martran marz marsdd marshut marx ipbsx ipbsy`) move only with tth at 83 or above, and tth goes below 83 only with martran at 1200 or above. A move from the GUI or a plan that would break this is refused and the SPEC log says why, for example `marz move refused: tth 1 is below 83`. The motor stays where it was, so a plan fails (`martran stopped at 1234, not 1250: 16000 steps off`) and the MCP `tweak` returns an error. The motor's record has a MAJOR alarm until the motor next moves, and its setpoint goes back to the readback, so the next ◀ or ▶ steps from where the motor is. SPEC's own moves (a macro, a command) cannot be refused: once one breaks the rule, spec-ioc sends SPEC a ^C, which stops every motor, and logs `interlock: …; sent ^C`. tth then stops a little below 83; move tth up before moving the mar.

Motors



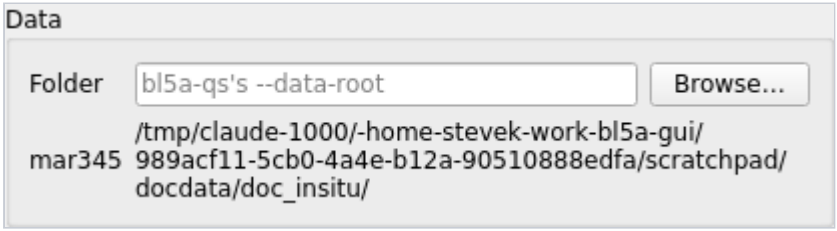
The motors panel

One row per motor the plans move: `sy shold th tth nu sx phi scell martran marsdd marshut`.

- **position** is the readback, with the record's precision and unit.
- An orange dot after it means the motor is moving.
- **step** is the tweak step (the motor record's `TWV`). Type a positive number and leave the field (Tab or click elsewhere) to set it.
- ◀ moves the motor by −step, ▶ by +step. spec-ioc turns each into a SPEC move.

Align position and **Measure position** stay disabled until BL5A gives these positions.

Data



The Data box

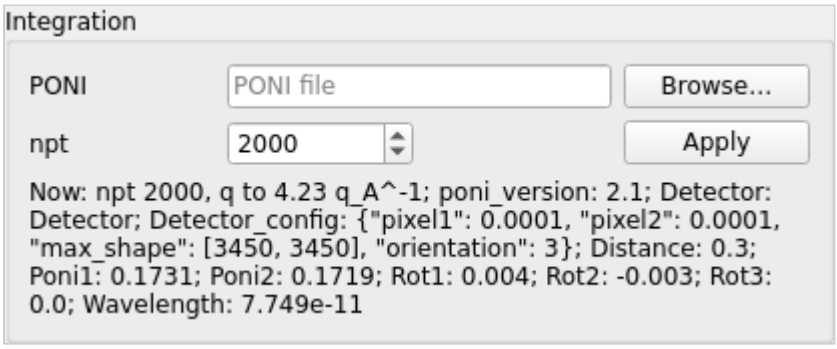
Everything an item records goes in one folder, `<data root>/<series>/`: its frames, its scans as the SPEC file `<series>.spec` and its integrated stacks. bl5a-qs makes the folder when the item starts. The series is the one typed in the form, or the plan's name when there is none, so the items of one series share a folder. Type the same series in **Alignment** and on the **Battery** form and a cell's alignment scans and frames land together:

```
<data root>/
  cell7/
    cell7.spec              the alignment scans
    cell7_0001.mar3450 …    the frames
    cell7.h5                the stack
  film_A/
    film_A.spec             each in-situ row's alignment scans
    filmA_30_0001.mar3450 …
    film_A.h5
```

- **Folder** is the data root of the items queued from now on; empty means bl5a-qs's `--data-root`. **Browse…** picks it, and the GUI remembers it. It must be an absolute path without spaces that is the same on the bl5a-qs, mar345 IOC and integrator hosts (see install.md).
- Each queued item carries the folder as it was when queued, so a change does not move items already in the queue. The queue's **Folder** column shows where each item goes.
- **mar345** is the folder the mar345 IOC writes to now (`FilePath_RBV`); bl5a-qs sets it before each frame. A red "missing" after it means the IOC does not see that folder (`FilePathExists_RBV` reads No).

A series is one folder name without spaces, `/` or control characters, and not `.` or `..`; the whole folder, `<data root>/<series>`, is at most 200 bytes. The mar345 detector server takes a frame's path as one word, so a space would split it. The forms refuse any other series, and queueing refuses a **Folder** with a space or a path that is too long. In the stack's file name, a character other than a letter, digit, `_`, `-` or `.` is written `%XX`: series `LiFePO4(1C)` gives `LiFePO4%281C%29.h5`. An item whose data root does not exist fails when it starts, with "data folder: … the data root … does not exist".

Integration



The Integration box

The text under the buttons is what the integrator integrates with now: npt, the q range and the PONI.

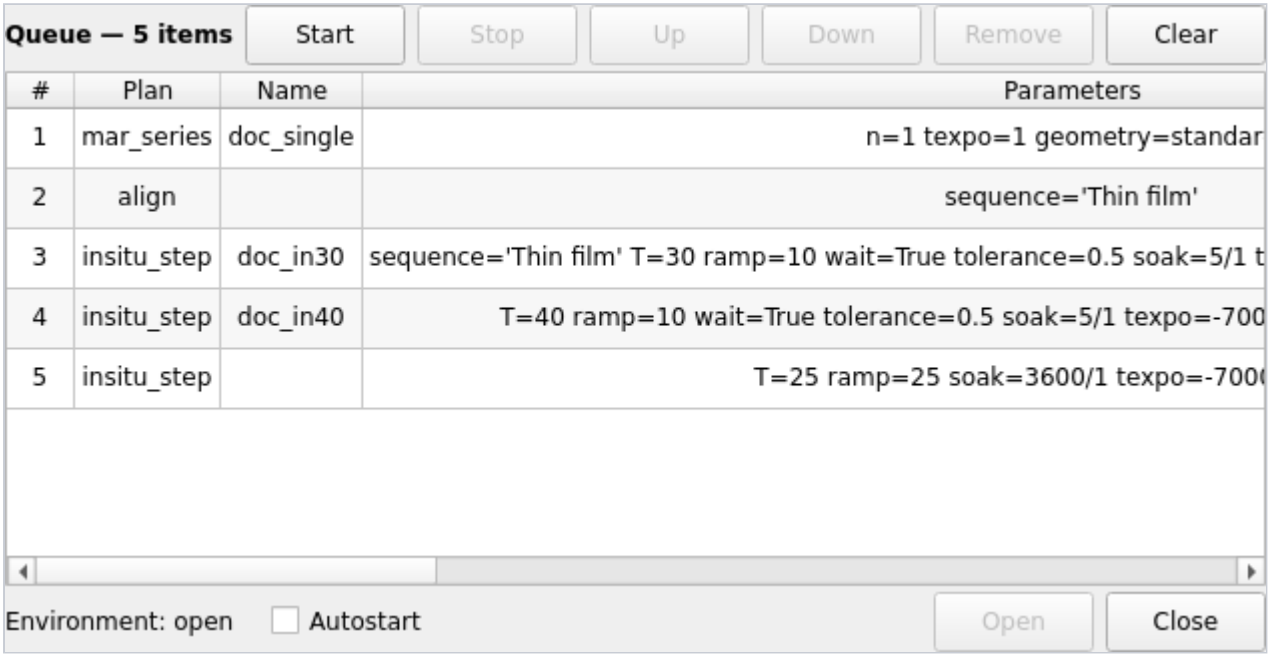
Control	What it is
PONI	a pyFAI PONI file for the full 345 mm plate; **Browse…** picks it
npt	the bins from q = 0 to the plate's far rim (default 2000)
Apply	reads the file and sends its text and npt to the integrator

Save and apply… on the Calibration tab puts the refined PONI's file in PONI and applies it the same way.

The integrator checks them first: a PONI it cannot use is refused, and the box adds "Refused:" and the reason. An accepted one applies to the frames after those already waiting to be integrated; until then the box shows it as "Next, after the queued frames". When it takes effect, Messages says "integrator: npt …, q to … from now on" and the Image tab's rings move to the new PONI.

A new npt, or a PONI that moves the q range, starts a new stack: the next frame of `cell7` goes to `cell7~2.h5`, and `cell7.h5` keeps its q axis. The Stack tab lists both. The integrator keeps the setting until it restarts, and then starts again from its `--poni` and `--npt`.

Queue



The queue with five items

The top row, marked ▶ on a green background, is the running item; the rest wait in order.

Column	Shows
#	the place in the queue
Plan	`mar_series`, `align`, `insitu_step` or `scan`
Name	the frame file name, when the plan takes frames
Parameters	the item's arguments, for example `sequence='Thin film' T=30 ramp=10 wait=True tolerance=0.5 soak=5/1 texpo=-70000 th_list=-0.05/0.11/0.13/0.15`
Folder	where the item's data goes: `<data root>/<series>`, or the series alone when the item goes to bl5a-qs's `--data-root`
User	the `--user` that queued it

Button	What it does
Start	runs the queue from its first item (only with items and nothing running)
Stop	stops the queue after the running item; that item is not interrupted
Up, Down	moves the selected item one place
Remove	takes the selected items out (also the Delete key); not the running item

Button	What it does
Clear	empties the queue after asking; the running item goes on

The row under the list belongs to the queue server:

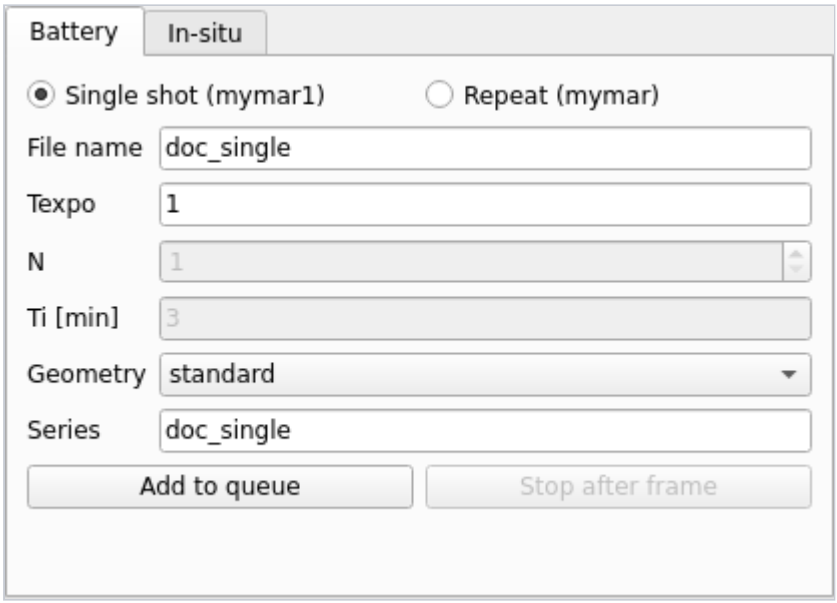
- **Environment: open / closed / not known.** Nothing runs while it is closed. **Open** opens it. **Close** is offered only while idle.
- **Autostart.** When ticked, what is added to the queue runs without a Start. The server unticks it when the queue ends in any way other than running dry: a failed or aborted plan, or a Stop. Tick it again here.

A click on a form's queue button sends one request. The rows of the In-situ table go in as one request, in table order, so nothing can slip in between them.

bl5a-qs checks an item's arguments only when the item runs. The forms check the same rules before they queue, so a bad value is caught at the form and not halfway through the night.

Battery measurements

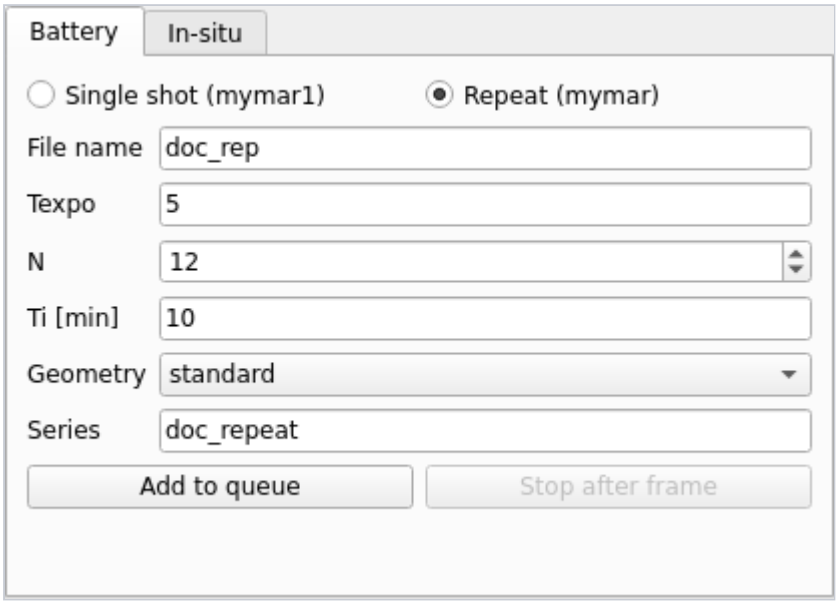
The Battery form queues `mar_series`: SPEC's `mymar1` (one frame) or `mymar` (a frame every Ti minutes).



Single shot

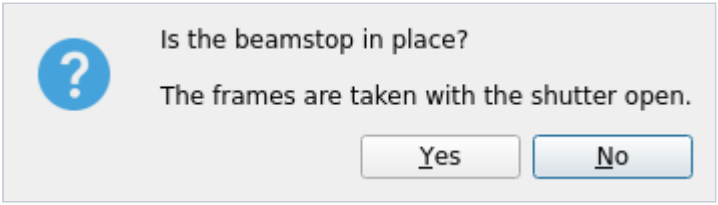
Field	Rule	Notes
Single shot / Repeat		Single shot is `mymar1`; Repeat is `mymar`
File name	1 to 27 characters, no spaces or quotes	the mar345 file name
Texpo	seconds 1 to 20, or a negative number of monitor counts	default 2
N	whole number, at least 1	Repeat only: the number of frames
Ti [min]	more than 2.5	Repeat only: from one frame's start to the next

Field	Rule	Notes
Scan mode	`static`, `rocking1`, `rocking2`	the macro each frame is taken with: `mar345expose` (still), `mar345_lab6` (th and phi rock from −1° to +1°), `mar345_lab6_wide` (−5° to +5°)
Filters	four blades, `0` or `1` each	SPEC's `fi` before the frames: the PFCU0 filters, default `0 1 0 0`
Series	a folder name (Data)	the data folder and stack the frames go to; empty means the file name



Repeat: twelve frames, ten minutes apart

Add to queue checks the fields first. A bad value opens a "Battery" box that names the field and the rule. Then the GUI asks:



The beamstop question

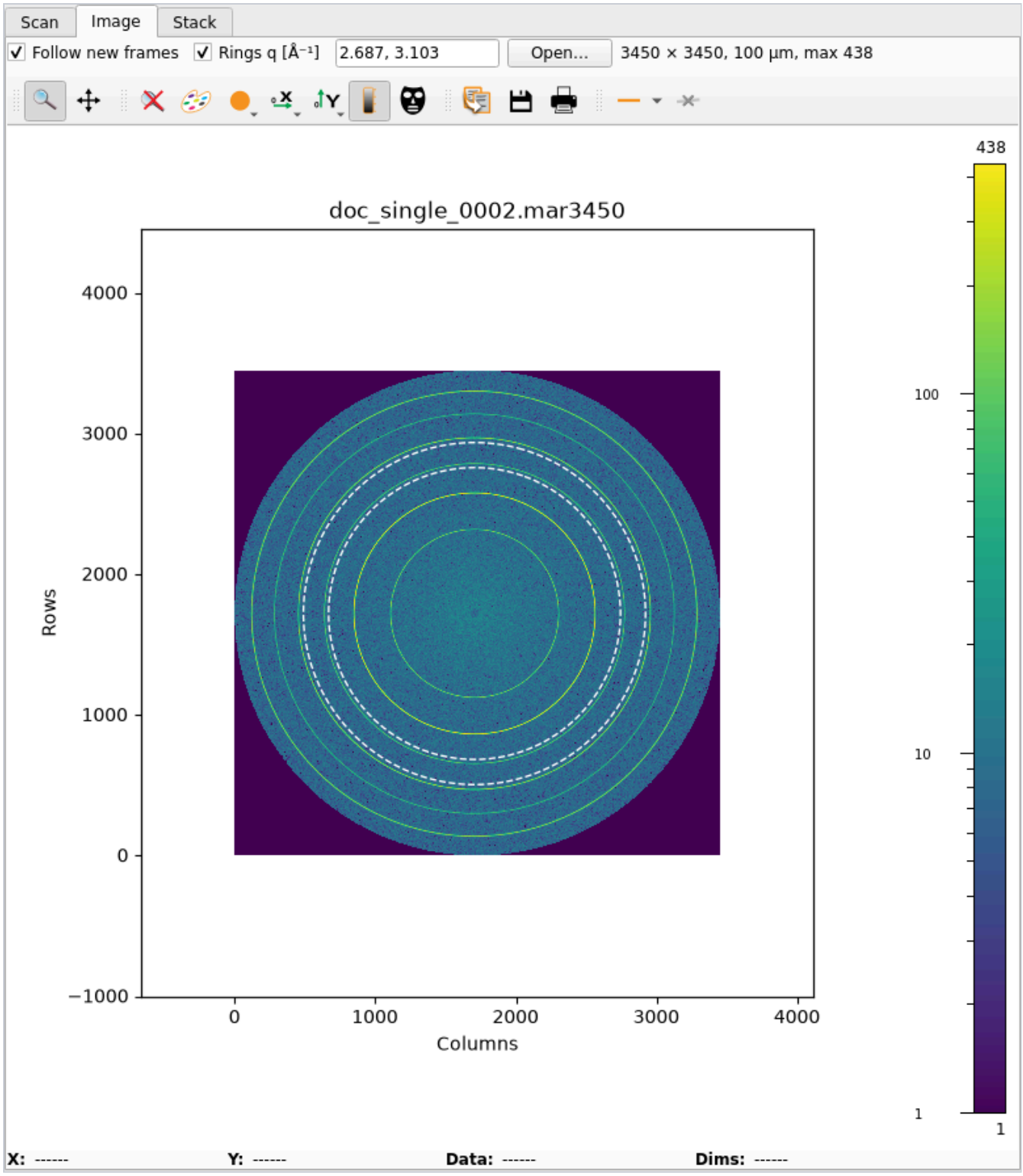
Yes queues the item; **No** queues nothing.

Set the geometry first. `mar_series` moves neither tth nor the mar's motors: the frames are taken where they stand.

What a `mar_series` does, in order:

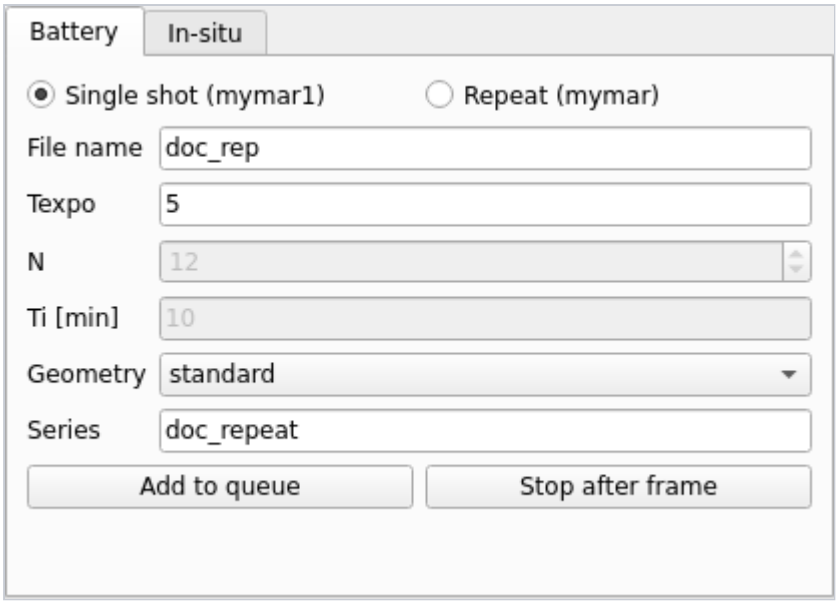
1. `mar345closeshutter`, `scell` to 0, `counters 2 3`, `fi <filters>`, `ic_range 4`, a 2 s wait, `scell` to 0.
2. For each frame: the item's data folder goes to the mar345 IOC's FilePath, then the scan mode's macro (`mar345expose`, `mar345_lab6` or `mar345_lab6_wide`) with `("<name>", <texpo>)` names the file, works the shutter, rocks th and phi in the rocking modes, counts, logs `exp_condition`, has the plate read out and runs `mar345readout`.
3. Frame *i* starts *i*·Ti after the first frame. Between frames the plan only waits.

The mar345 writes `<name>_<number>.mar3450` in the item's data folder. The number is the IOC's file counter, which runs on from one name to the next.



A single shot on the Image tab

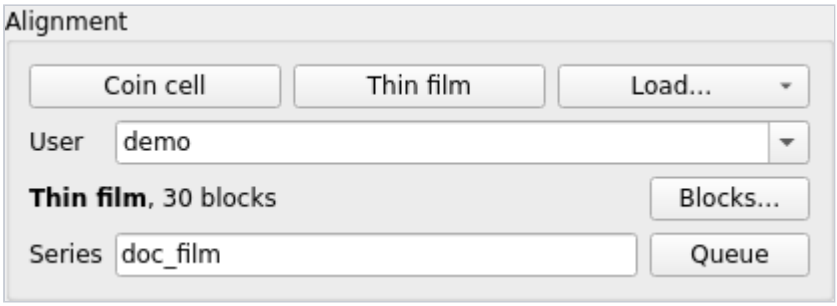
Each frame appears on the Image tab once it is read out, and is added to its series on the Stack tab once the integrator has integrated it.



While a series runs

Stop after frame is live while a `mar_series` runs. It lets the frame being taken finish, then ends the series: it asks for a pause that takes effect between frames, and sends Stop once the plan has paused. The status line says "stopping after the frame" in the meantime.

Alignment



The Alignment box

An alignment is a sequence of blocks run in order. Two sequences come with the GUI and are read-only: **Coin cell**, SPEC's `coin_align`, and **Thin film**, SPEC's `align_in.mac`. Each user's own sequences are kept in the Data folder, one file each, `<data folder>/alignments/<user>/<name>.json`.

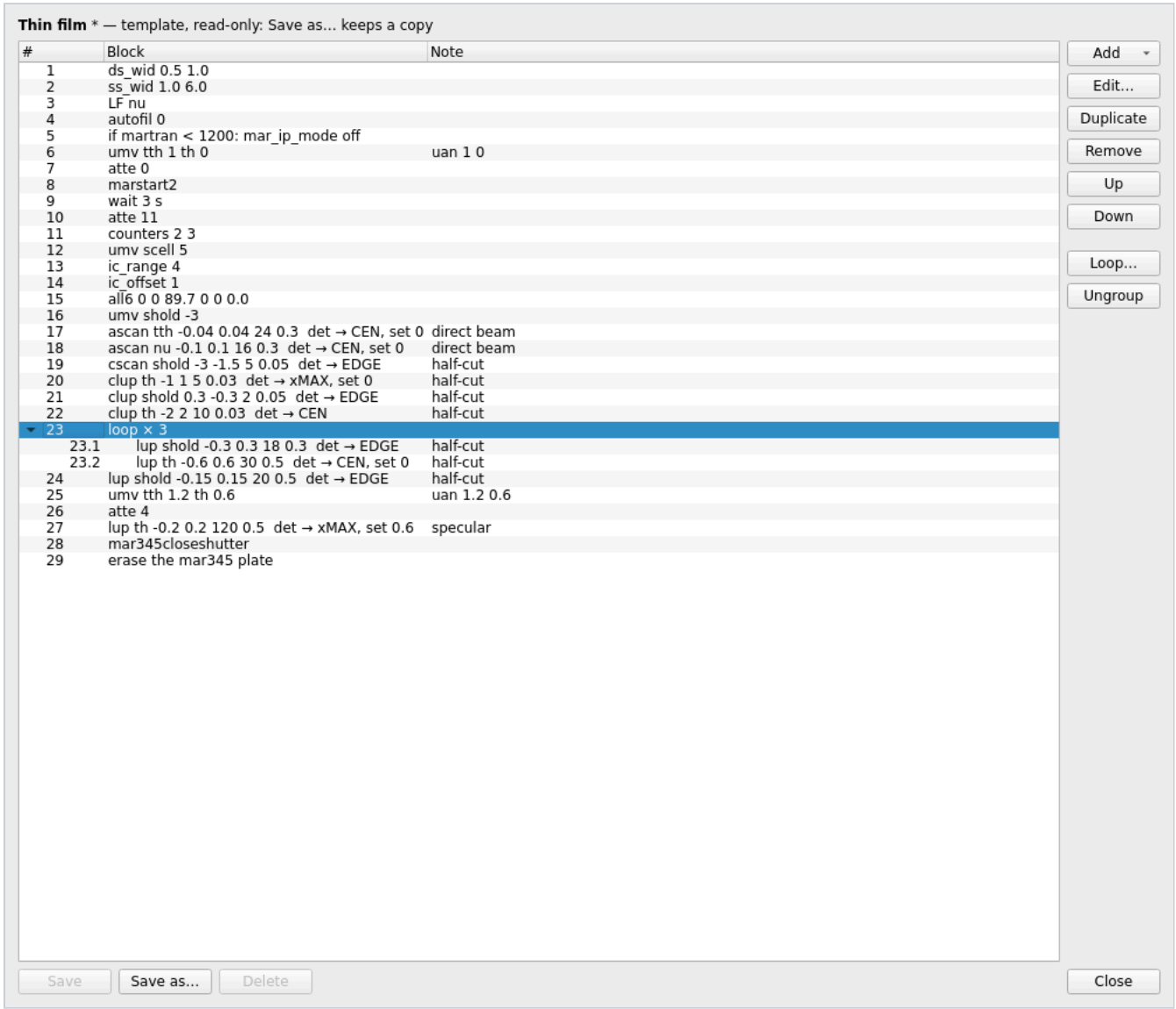
The box names the sequence last loaded and how many blocks it has; the GUI starts with Thin film. Its blocks are edited in a window of their own, Blocks.

Field	What it does
Coin cell, Thin film	loads that sequence's blocks and opens the Blocks window
Load…	the same for one of the User's saved sequences, picked from its menu
User	whose sequences Load…, Save and Delete use; type a name, or pick one that has saved sequences; empty: `bl5a`
Blocks…	opens the Blocks window again
Series	groups the runs in the alignment history, is stored with the run and names the data folder its scans go to; empty means each run stands on its own, its scans in the folder `align`

Field	What it does
Queue	queues the plan `align` with the blocks loaded, changed or not

The name has `*` after it while the blocks have unsaved changes. Queued so, the run records the name with "(edited)", for example `Thin film (edited)`. The item carries the blocks, so a change made after queueing does not change the queued item. Loading another sequence, or closing the GUI, asks before dropping unsaved changes. Every alignment ends by recording sx, sy, shold, th and phi; the history shows sy and shold.

The Blocks window



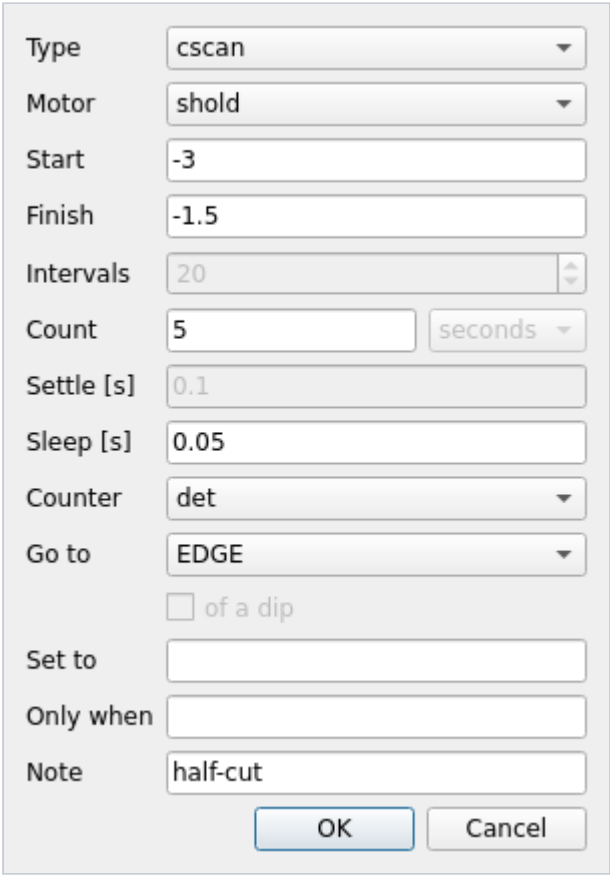
The Blocks window

Each row is one block: its number (#), then the SPEC line it sends, or the scan it runs as the Stages table names it with the counter and where the motor goes, then its note. A loop's blocks are under it, indented and numbered inside it: 4.2 is block 2 of the loop block 4, as an error names it ("block 4 (loop) block 2 (scan): …"). Closing the window keeps the blocks loaded.

Select a row, or neighbouring rows in one list (Shift-click); a selected loop counts with the blocks inside it. Then:

Button	What it does
Add	a new block of the kind picked from its menu, after the selected ones, in their list
Edit…	the current block's form (or double-click the row)
Duplicate, Remove	copies or removes the selected blocks; a loop left with none goes too
Up, Down	moves them one place within their list
Loop…	puts the selected blocks into one loop block that runs them in order, the number of times asked
Ungroup	puts the selected loop's blocks in its place, run once; its times, Only when and note go
Save	writes the sequence under its name; not for Coin cell or Thin film
Save as…	writes it as the user's under a new name; asks before replacing one
Delete	deletes the user's saved sequence and loads Thin film
Close	closes the window

Saving needs a Data folder. A name is 1 to 64 characters, not starting with `.`, without `/` or `\`, and not Coin cell or Thin film.



A scan block

Block	Fields	What it does
scan	type, motor, start, finish; count (ascan, lup: seconds per point, or monitor counts on icxrd, SPEC's MON, which SPEC shows as a negative count time; cscan, clup: seconds of the whole move); intervals and settle (ascan, lup) or sleep (cscan, clup); counter; go to; of a dip; set to	scans, checks the scan (Before a stage moves), moves the motor to the statistic picked in go to and, when set to has a value, redefines that position with `set`; ascan and cscan are absolute, lup and clup relative to where the motor is
move	motor position pairs, `tth 1 th 0`	`umv` of them together
spec	one command line	sends it to SPEC
wait	seconds	waits
erase		erases the mar345 plate
loop	times	runs its blocks in order, that many times over; made with Loop…, its blocks edited in the list; loops may hold loops

Any block can also have **Only when**, for example `martran < 1200`: the block runs only while the motor is below (`<`) or above (`>`) the value; a loop's is checked once, before it starts. **OK** checks the block as bl5a-qs will and names the field that breaks a rule. Each scan block is one row of Stages.

Coin cell

SPEC's `coin_align`:

1. `marshut` to 0, `mon1_setup 1050 100 300 3.0 1.0`, `setkappa 50`, `counters 2 5`, `fi 1 0 0 0`, `fil 0 0 0 0`.
2. A fly scan of sy from −2.5 to 2.5 on ic1; sy goes to pl_COM.
3. A fly scan of shold from −2 to 2; shold goes to pl_COM.
4. `mar345closeshutter`, the plate is erased, `marshut` to 340.

`coin_align` ends with `mar_ip` without arguments, which only prints its usage in SPEC, so the sequence does not send it.

Thin film

SPEC's `align_in.mac`. It sets the slits, attenuators and counters (`mar_ip_mode off` only when martran is below 1200), moves to the direct beam (`uan 1 0`, `all6 0 0 89.7 0 0 0.0`, `shold` to −3) and then runs these stages. Each stage scans, moves the motor to the picked statistic and, where a value is given, redefines that position with `set`.

Stage	Pick	Then
tth −0.04 to 0.04	CEN	`set tth 0`
nu −0.1 to 0.1	CEN	`set nu 0`
shold −3 to −1.5, fly	EDGE	
th −1 to 1 relative, fly	pl_xMAX	`set th 0`
shold 0.3 to −0.3 relative, fly	EDGE	
th −2 to 2 relative, fly	CEN	
shold −0.3 to 0.3 relative	EDGE	
th −0.6 to 0.6 relative	CEN	`set th 0`

Stage	Pick	Then
shold −0.15 to 0.15 relative	EDGE	
`uan 1.2 0.6`, `atte 4`, then th −0.2 to 0.2 relative, 120 intervals	pl_xMAX	`set th 0.6`

It ends with `mar345closeshutter` and an erase of the plate.

EDGE is where a shold scan falls through halfway between its maximum and its floor, the height at which the sample takes half the beam. The floor is the scan's minimum when at least 3 of the 25 bins lie within 3 times the noise of a difference of two bins of the lowest; a scan that ends on the slope shows no floor, and it is taken as zero. `align_in.mac` moves to CEN, which for an edge lies halfway between that point and the maximum, and on the open-beam plateau the maximum is wherever the noise puts it: over four scans of one edge on the beamline CEN moved by 0.24 while the half point stayed within 0.007.

A scan block with **of a dip** checked goes to CEN, COM or xMAX of a dip: of the scan turned over, its maximum less each point, so the dip's bottom is the maximum and the plateau around it zero. CEN is then the centre at half the dip's depth and xMAX the scan's minimum. EDGE has no dip form.

Before a stage moves

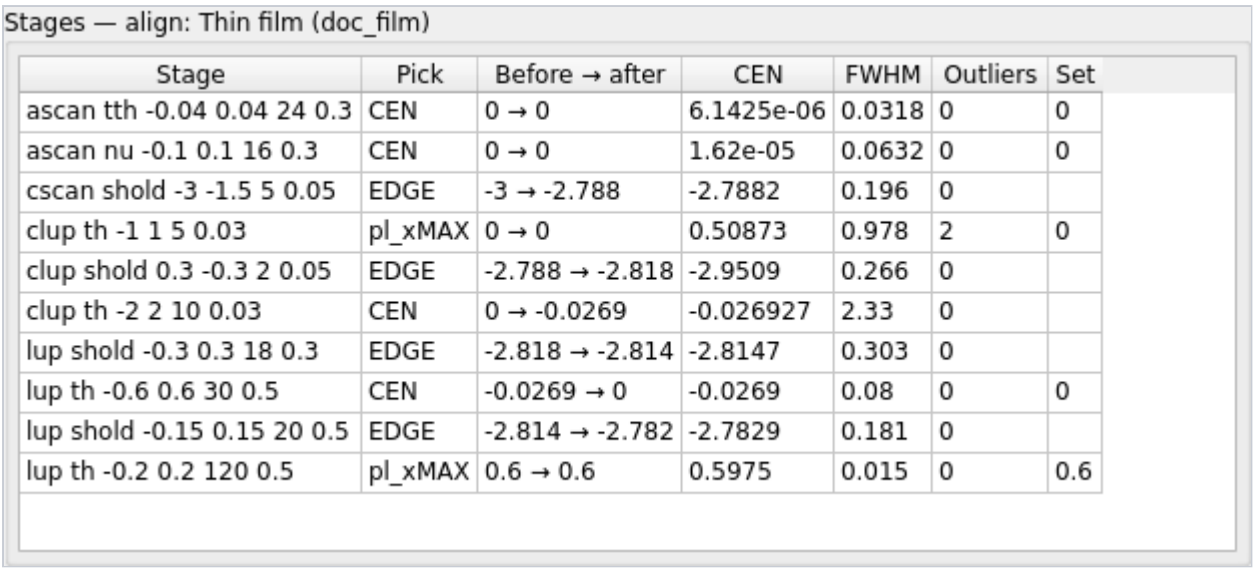
A point more than 3σ from the median of itself and the two points either side of it is replaced by that median, and so is a point that is not a number. σ is that of the five points, but at least the square root of their median and the noise of the whole scan. A cscan's second sample counts about 1.7 times as long as the others and is often replaced this way; left in, it can be the maximum CEN and pl_xMAX follow. The statistics are those of the filtered points.

The stage then puts the motor back where it was before the scan and ends the plan, which fails with the reason Messages shows, when

- the scan shows nothing: averaged into 25 bins, it changes by less than 10% of its largest bin, or by less than 5 times the noise of a bin;
- a peak's maximum is the scan's first or last point, or both of its neighbours are at or below half of it (a peak one point wide); for a dip, the same of the scan turned over: the minimum is an end point, or the dip is one point wide at half its depth;
- the bins on either side of a peak's largest bin do not fall below it by more than 3 times the noise of a difference of two bins: the peak may lie past the end, with the end point's noise putting it below the one before;
- an edge never falls to half its maximum, or falls to half on both sides. A step that does not fall to half, such as a sample that takes a fifth of the beam, is not taken for an edge over a high floor.

A peak that never falls to half inside the scan, such as a rocking curve wider than a th scan, still goes to CEN, which is then the maximum.

Stages


Stages — align: Thin film (doc_film)

Stage	Pick	Before → after	CEN	FWHM	Outliers	Set
ascan tth -0.04 0.04 24 0.3	CEN	0 → 0	6.1425e-06	0.0318	0	0
ascan nu -0.1 0.1 16 0.3	CEN	0 → 0	1.62e-05	0.0632	0	0
cscan shold -3 -1.5 5 0.05	EDGE	-3 → -2.788	-2.7882	0.196	0	
clup th -1 1 5 0.03	pl_xMAX	0 → 0	0.50873	0.978	2	0
clup shold 0.3 -0.3 2 0.05	EDGE	-2.788 → -2.818	-2.9509	0.266	0	
clup th -2 2 10 0.03	CEN	0 → -0.0269	-0.026927	2.33	0	
lup shold -0.3 0.3 18 0.3	EDGE	-2.818 → -2.814	-2.8147	0.303	0	
lup th -0.6 0.6 30 0.5	CEN	-0.0269 → 0	-0.0269	0.08	0	0
lup shold -0.15 0.15 20 0.5	EDGE	-2.814 → -2.782	-2.7829	0.181	0	
lup th -0.2 0.2 120 0.5	pl_xMAX	0.6 → 0.6	0.5975	0.015	0	0.6

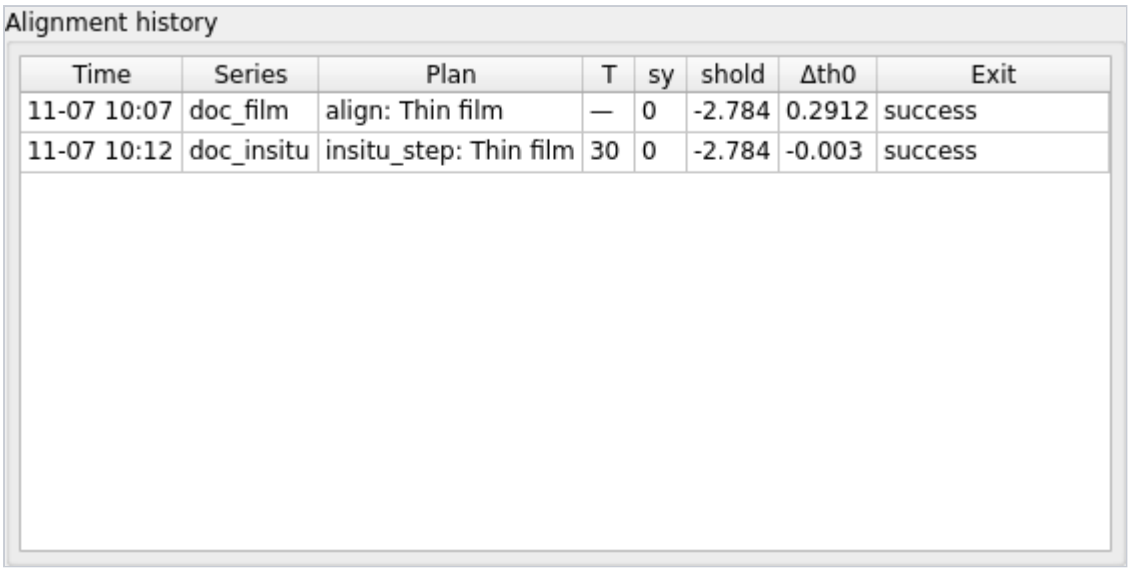
The stages of a Thin film run

The Stages table shows each stage of the alignment running now. Its title names the plan, the sequence and the series. It is cleared when the next `align` or `insitu_step` starts, including an `insitu_step` row without alignment.

Column	Shows	Tooltip
Stage	the scan, as a SPEC command line	
Pick	the statistic the motor was moved to: CEN, pl_COM, pl_xMAX or EDGE, followed by `dip` for a dip	the target position
Before → after	the motor before the scan, and after the move and any `set`	
CEN	the centre at half maximum	COM and the position of the maximum
FWHM	the full width at half maximum	
Outliers	how many points the outlier filter replaced	
Set	the value given to `set`, empty when the stage did not set	how far the user offset moved

The statistics are splot's (`CEN`, `pl_COM`, `pl_xMAX`), the ones SPEC's macros read, of the points after the outlier filter (see Before a stage moves); a dip's are of those points turned over.

Alignment history



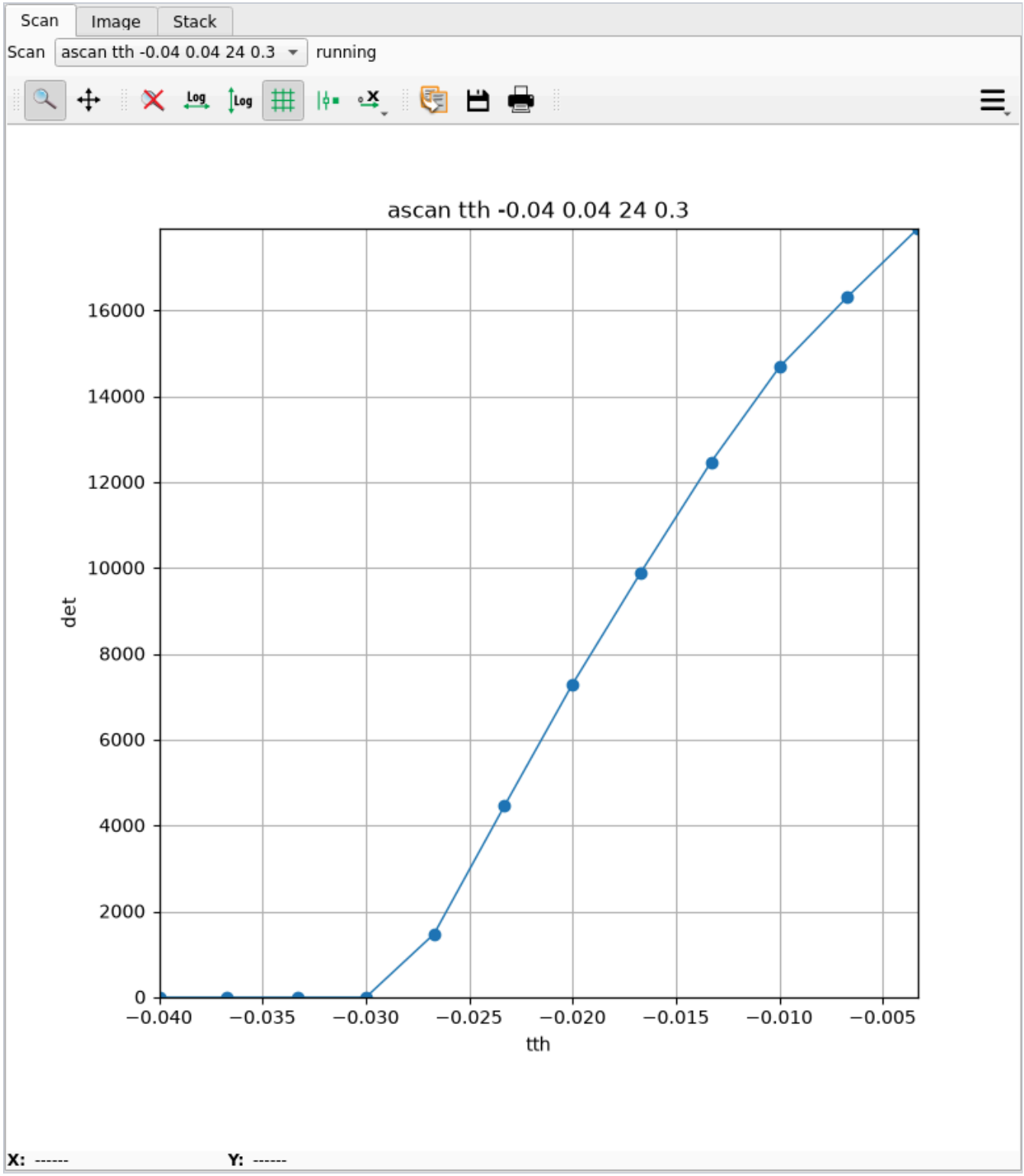
The alignment history

One row per finished run that aligned:

Column	Shows
Time	when the run started
Series	the series it was queued in
Plan	`align` or `insitu_step`, with the sequence it ran
T	the `insitu_step` setpoint, °C
sy, shold	where the run left them
Δth0	how far this run moved th's zero
Exit	`success`, `fail` or `abort`; its tooltip gives the reason a run did not succeed

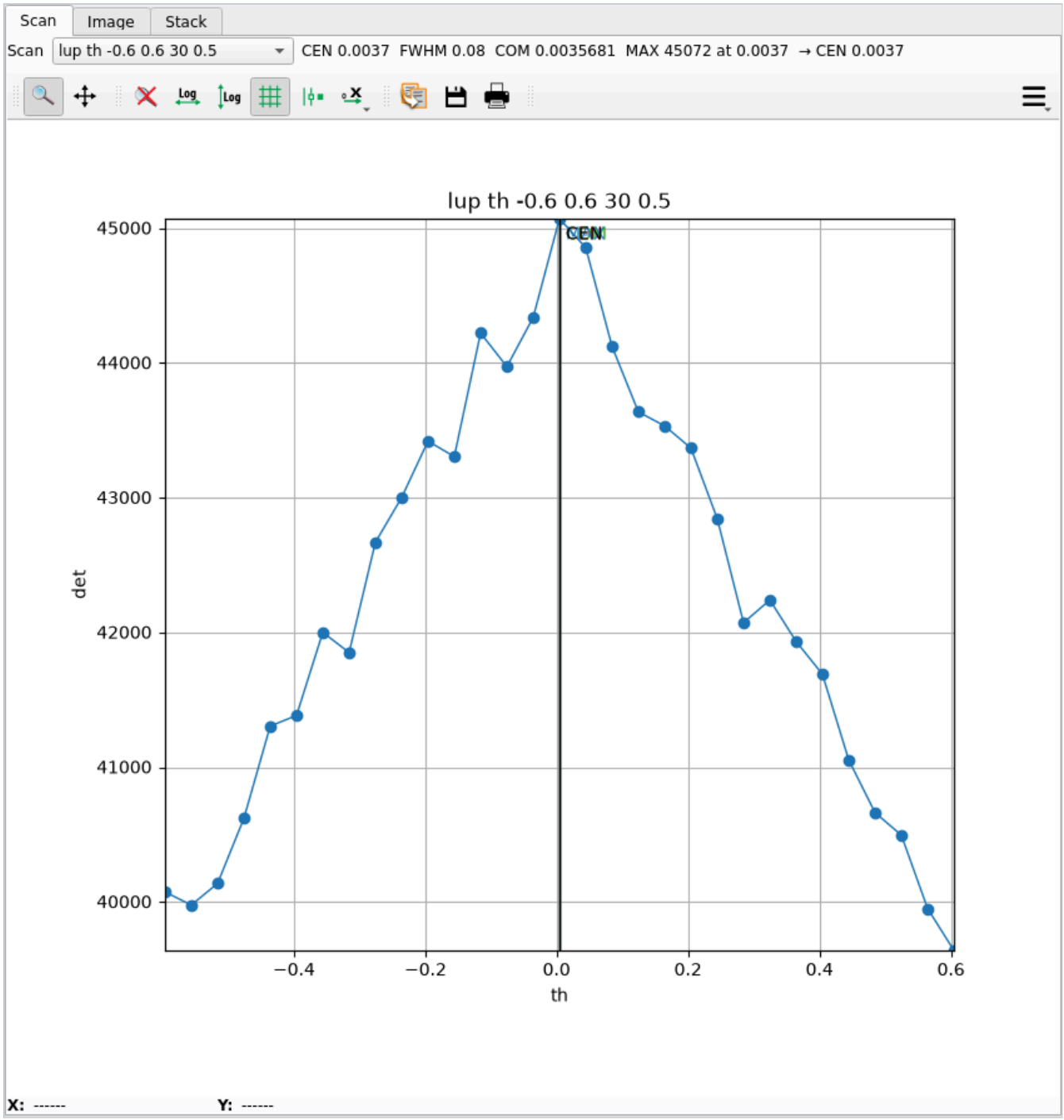
Δth0 is the sum of this run's `set th` corrections, not th0 itself: SPEC's dial position is not published, so the absolute offset is not known to the GUI.

Alignment tab



A scan being drawn

A scan is drawn point by point as it runs, at most five times a second. A fly scan runs in SPEC, which keeps its sampling timing; each of its points is drawn as SPEC takes it (spec-ioc passes on SPEC's `SCAN_PT` events), and is the counts SPEC took in one sample, at the middle of the motor's travel over it. When SPEC finishes, its full table of the scan adds any point not seen yet; a table that does not begin with the points already drawn fails the scan.



A finished scan with its statistics

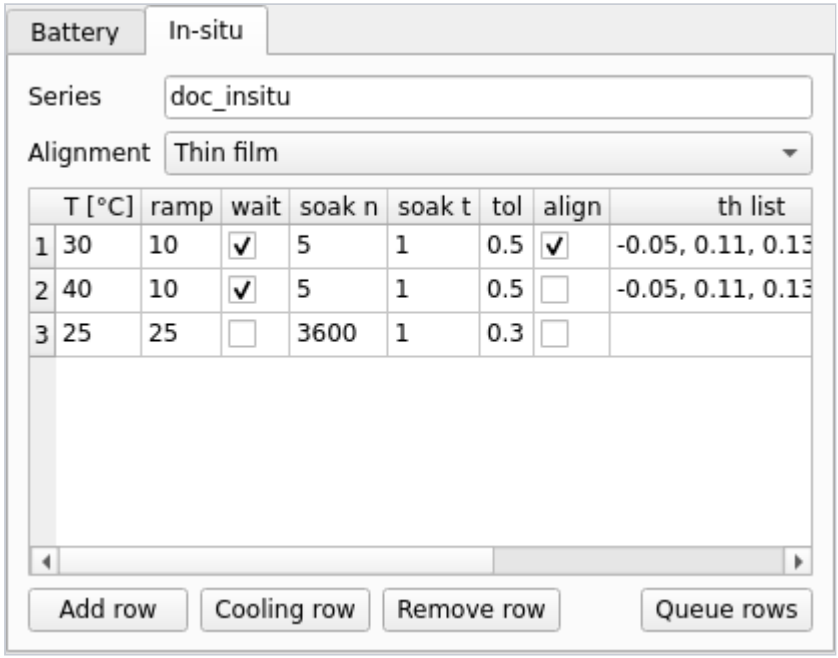
When the scan ends, the line above the plot gives CEN, FWHM, COM and the maximum and its position, and markers are drawn: CEN red, COM green, MAX blue, and the stage's target black with the pick's name. In the picture the stage picked CEN, so the black target lies on the red CEN line. A stage's scan shows the stage's statistics, of the points after the outlier filter, and how many points the filter replaced; the curve is the points as taken, so where a point was replaced the markers can lie off it. A dip's stage gives DIP, its depth, and its position for the maximum, and marks the minimum MIN. Any other scan shows the statistics of its points as taken. A scan that did not end in success shows its exit status and why; the whole line is in its tooltip when it does not fit.

The **Scan** list keeps this window's last 100 scans; pick one to look at it again. Every scan, from any window or none, is also appended to the SPEC-format file of its item's data folder, `<series>.spec` (see Data and install.md section 4.5), for splot or PyMCA. The toolbar is silx's: zoom, pan, log axes, grid, save, print.

The **Scan** form, on the left beside Battery and In-situ, queues one scan as a `scan` item. Its boxes are an alignment scan block's (see The Blocks window): Type `ascan` or `cscan` (positions; the motor stays at finish) or `lup` or `clup` (offsets from where the motor is, and it goes back there after), a step scan's intervals, count and settle, a fly scan's sweep time and sleep, and the Counter whose statistics are taken. **Go to** `none` leaves the motor where the scan left it; CEN, COM, xMAX or EDGE then moves it there, as an alignment stage does, after the same outlier filter, with **of a dip** and **Set to** as in a block. The scan is drawn here, its target marked, and goes to its series' `.spec` file like an alignment's scans; with no Series it goes to the data folder `scan`. A scan item's move is not an alignment stage: it is not in the Stages table or the alignment history.

In-situ (film against temperature)

The In-situ form is SPEC's `myinsitu` as a table. Each row is one `insitu_step`; the rows run in order and their frames go to one series.



The in-situ table: 30 °C, 40 °C, then the cooling row

Column	Default	What it is
T [°C]	empty	the setpoint (`settemp T`); empty leaves the temperature alone
ramp	empty	°C/min (`set_ramp`)
wait	off	wait until T is reached (`changec T` instead of `settemp T`)
soak n	empty	`loopscan` points while soaking
soak t	empty	seconds per soak point
tol	0.3	with wait: how close to T counts as reached, °C (`changec`'s `DELTA_K`)
align	on	run the Alignment sequence before the frames
th list	-0.05, 0.11, 0.13, 0.15	one frame at each th, comma separated; empty takes no frames
texpo	-70000	seconds (> 0) or monitor counts (< 0)
name	empty	the frames' file name

Series names the data folder and stack of every row; empty means the first row's name. **Alignment** is the sequence each row with align runs: Thin film, or one of the Alignment box's user's sequences.

One row runs, in order:

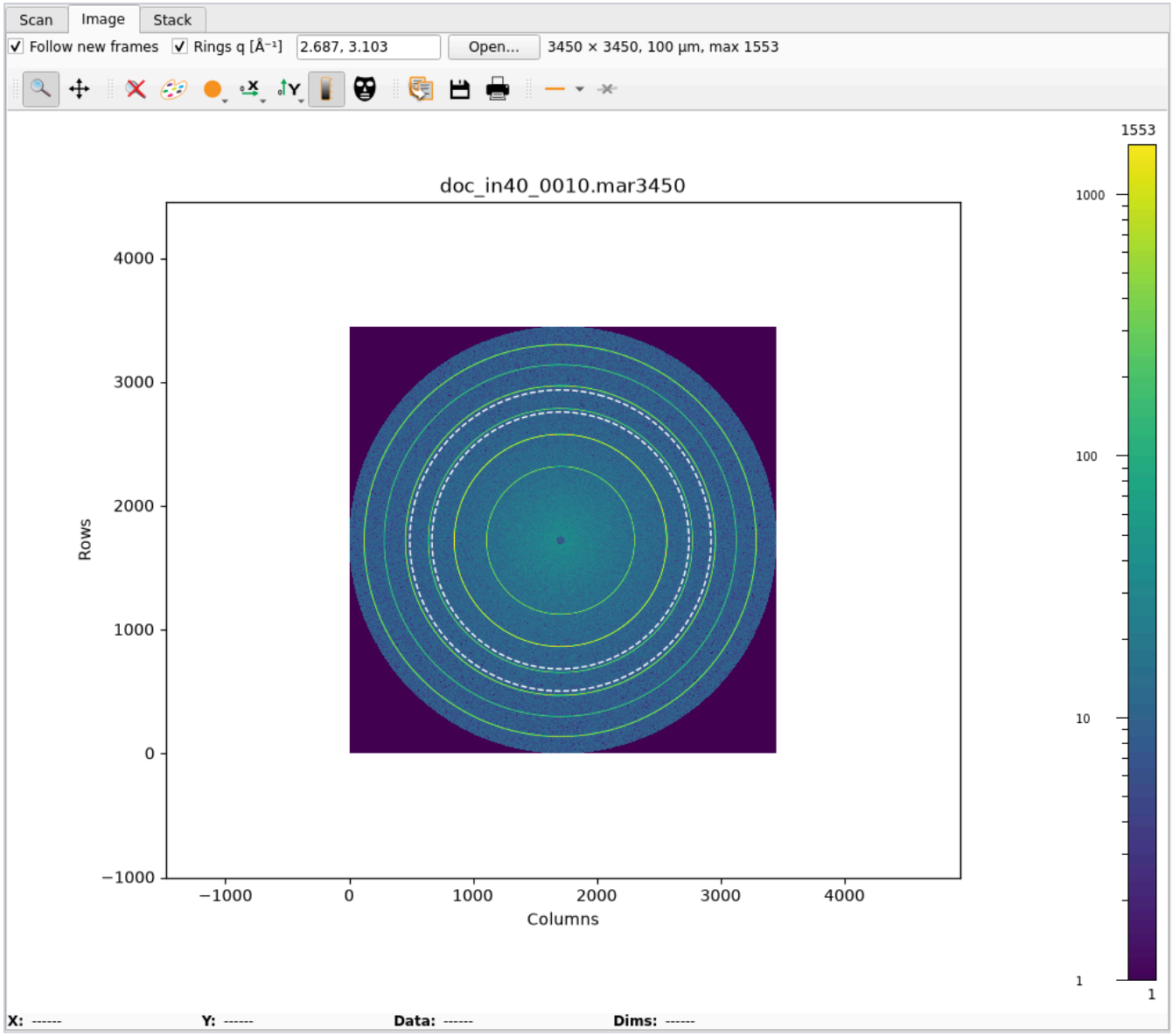
1. Temperature, only the parts filled in: `qdo` of the beamline's `temperature.mac`, `set_ramp <ramp>`, then `settemp T`, or with wait `changec T` with `DELTA_K` set to tol, then `loopscan n t`.
2. With align: the Alignment sequence (Thin film unless another is picked).
3. With a th list: `mymar_in`. If tth is below 40 it first closes the shutter, `uan 45 0`, takes tth to 83 and then the mar345 to the in-situ positions. Then `uan 83 0`, `counters 2 3`, `fi 1 0 0 0`, `ic_range 5`, `ic_offset 0`, `mar_ip7 25 440 42.4 340 81.8 1.2 -6.8`, 5 s, `scell` to 0.9, a frame at each th, `ic_offset 1`, `fi 1 0 1 1`, the mar345 out of the way, tth to 42, and `uan 1 0`. The macro has 81 and moves tth with the mar345; the plan keeps to the mar345's interlock (Run buttons).

Button	What it does
Add row	an empty row with the defaults
Cooling row	`myinsitu`'s last step: ramp 25 °C/min to 25 °C, no wait, soak 3600 points of 1 s, no alignment, no frames
Remove row	removes the selected rows
Queue rows	checks every row, then queues them all in order

The checks: name 1 to 27 characters without spaces or quotes (only when the th list is not empty); texpo not 0; ramp above 0; wait needs a T; with wait, tol above 0 (without wait tol is not used); soak needs both n (a whole number, at least 1) and t (above 0); a row with no T, ramp or soak, no alignment and no th list does nothing and is refused. A bad row opens an "In-situ" box that names the row and the field; nothing is queued then.

The cooling row soaks for an hour. Queue it last.

Image tab

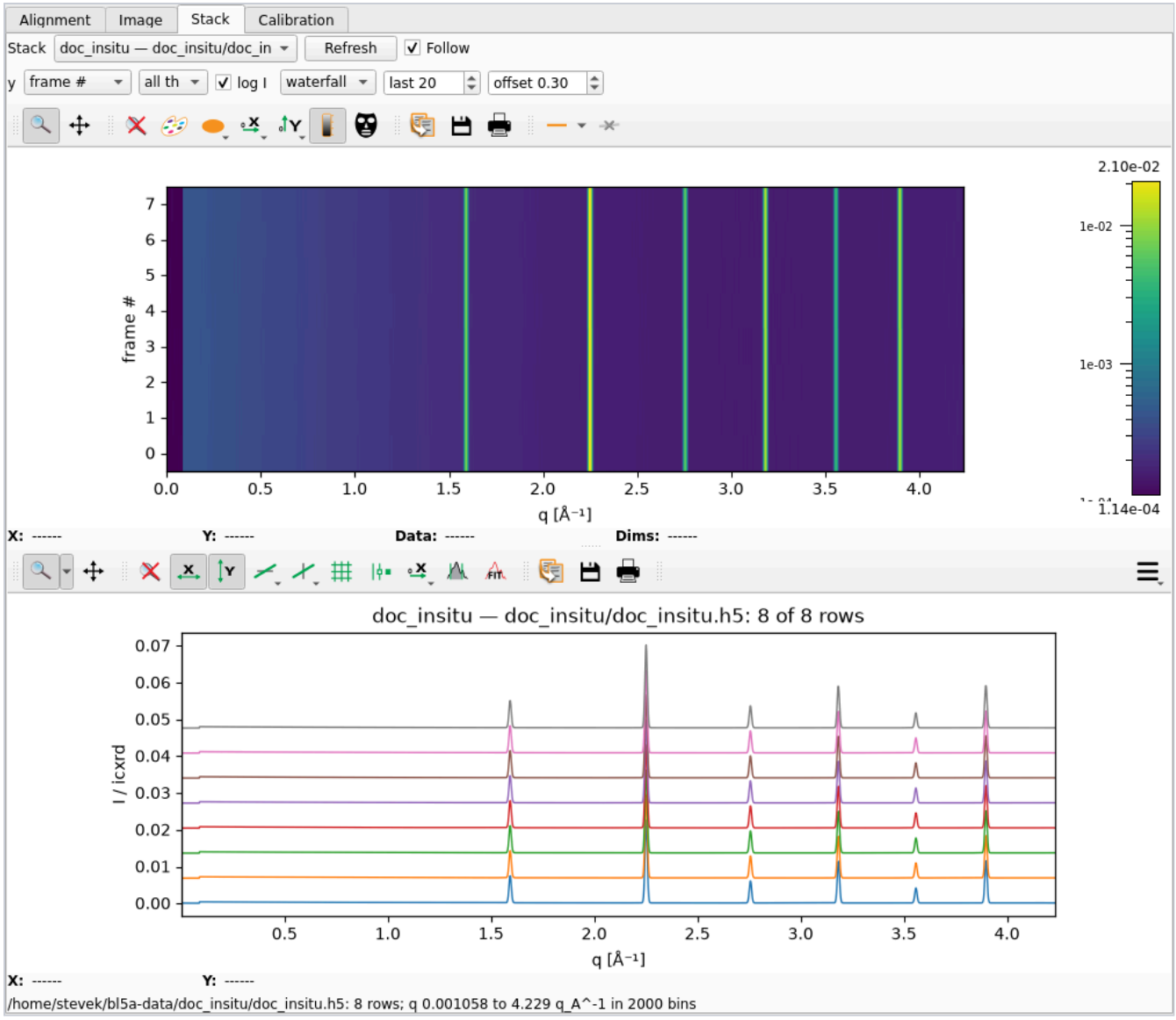


The last frame with the reference rings

- **Follow new frames**: each new frame is loaded as it is written. The file is read off the GUI's thread, from the path the mar345 IOC reports, so that path must be readable on the GUI's computer.
- The image is drawn on a log colour scale; zero counts are drawn as 1.
- **Rings q [Å⁻¹]**: comma-separated q values, drawn as dashed white circles where the integrator's geometry puts them. The defaults are Al 111 and Al 200 (2.687 and 3.103). The simulated film in the picture has no aluminium, so its rings fall elsewhere.
- **Open...** loads any mar345 file.
- The line after the buttons gives the image size, the pixel size and the largest count, or the reason a file could not be read.

The rings use the integrator's PONI, which the GUI asks the integrator for. The integrator uses one PONI for every frame, so the rings and the stacks are right only for the detector position that PONI was measured at (see install.md).

Stack tab



A series stacked: the heatmap and the waterfall

The integrator integrates every frame and appends it to its series' stack in the item's data folder. A stack is one series on one q axis (see Integration). The tab shows one stack:

- the heatmap: q across, one row per frame (or by time), log colour;
- the profiles: I / icxrd against q for the last N rows.

Control	What it does
Stack	every stack the integrator holds, as `series — folder/file`, in path order; at start the last one is shown
Refresh	asks the integrator again for its settings and its stacks
Follow	each new row brings up its stack (default on)
y	`frame #`, or `time [min]` from the first frame
all th / th v	all frames, or only those taken at one th
log I	log colour scale (default on)

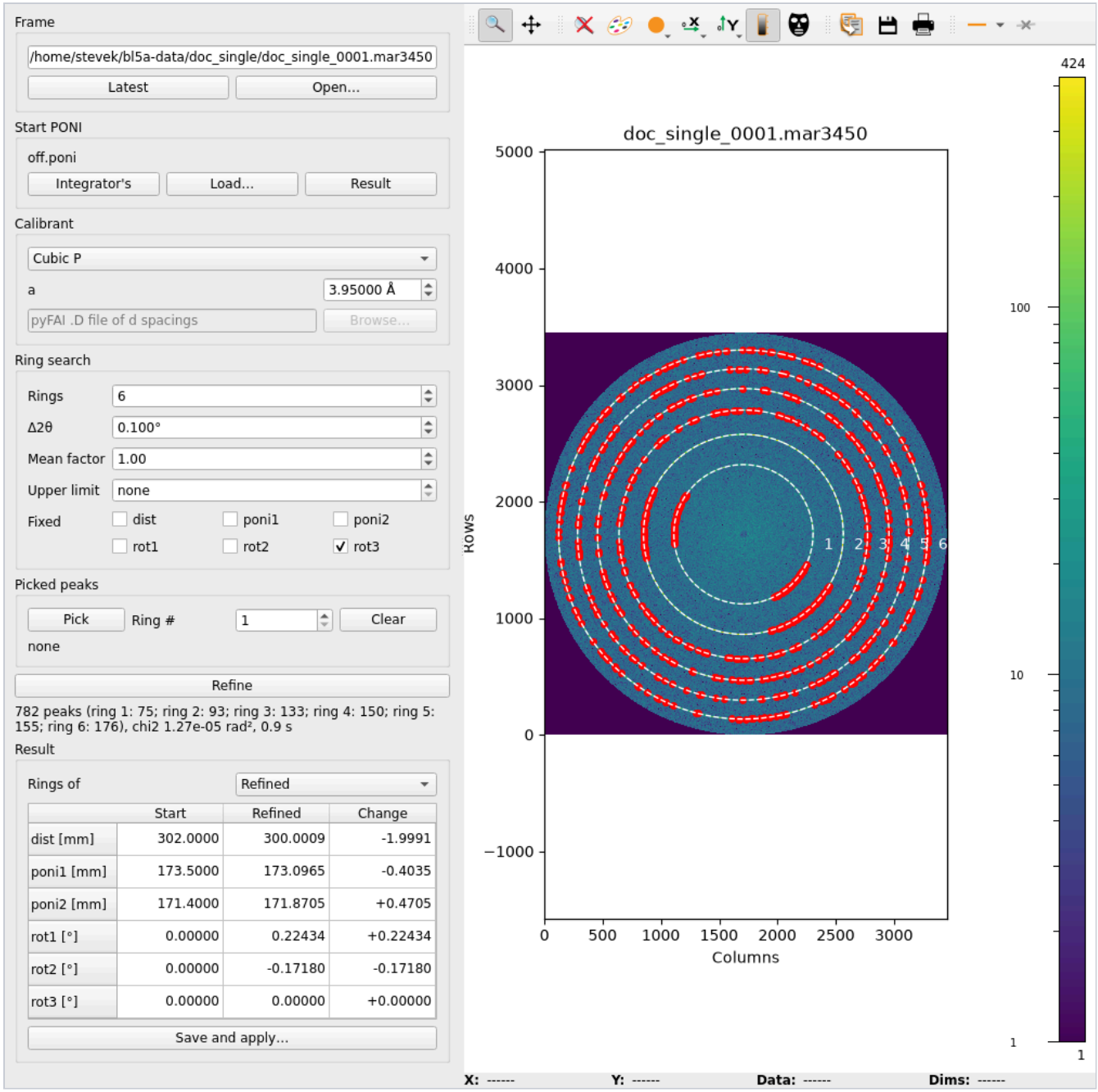
Control	What it does
overlay / waterfall	profiles drawn on top of each other, or stepped up
last N	how many of the newest rows the profiles show (default 20)
offset	the waterfall step, as a fraction of the rows' median maximum (default 0.3)

New rows arrive by themselves. A new stack shows up in the list when its first frame is integrated. With Follow on, the tab shows the stack of the newest row, new or listed; off, it stays on the stack shown, or shows the new one when none was shown.

The line at the bottom gives the stack's file, its number of rows and its q axis. It reads "integrator: not answering" when the integrator does not reply. The integrator's settings are in the Integration box.

The integrator reads a folder's stacks once, at its start or at the folder's first row. A stack file deleted on disk while it runs stays in the list until the series' next frame, which starts the file again.

Calibration tab



The simulated frame refined from a start 2 mm off

The tab refines the plate's PONI on a frame of a calibrant the way Dioptas's automatic refinement does it; the integrator does the work, with rsFAI. From the start PONI it searches the rings in turn, from the innermost: the pixels within Δ2θ of a ring that stand above Mean factor × the band's mean + its standard deviation give the ring's peaks, and the geometry is refined on every peak found so far before the next ring. Pixels off the scanned disc are left out.

Control	What it does
Latest, Open…	loads the mar345's last frame, or any mar345 file
Integrator's, Load…, Result	the start PONI: the integrator's, which it follows as the integrator's changes; a PONI file of the full 345 mm plate; or the last refined one, to refine again from
Calibrant	LaB6, CeO2, Si (NIST SRM 660c, 674b, 640c) or Al, with pyFAI's d spacings; Cubic P: every hkl of the lattice constant a; .D file: a pyFAI d-spacing file

Control	What it does
Rings	the rings searched, from the innermost (default 15); those the PONI's wavelength does not reach are left out
Δ2θ	the half width of each ring's band (default 0.1°)
Mean factor	default 1; Dioptas's 3 finds only the inner rings of the simulated frame, whose rings are broad
Upper limit	pixels above this count, saturated ones, are left out (default none)
Fixed	the parameters held at the start value; rot3 by default, which round rings barely show
Pick, Ring #, Clear	peaks picked by hand, below
Refine	sends the frame, the start PONI, the calibrant and the picks to the integrator
Rings of	the PONI the dashed rings are drawn with: Start, Refined, or none
Save and apply…	writes the refined PONI to a file and applies it as the Integration box's Apply does

The line under Refine gives the peaks found on each ring, chi2 (the sum of their squared 2θ misfits, rad²) and the seconds it took, or "Failed:" and the reason. The peaks are drawn as red dots, the rings move to the refined PONI, and the table gives the start, the refined value and the change.

A start too far off puts the bands beside the rings, and the search finds nothing or the wrong peaks. Then press **Pick**, set **Ring #** (1 the innermost) and click on that ring in a few places, then on another ring; each click takes the brightest pixel within 5 pixels, drawn as an orange ×. The integrator refines the start on the picks before the search. A second click within 0.4 s is a double click and picks nothing. **Clear** drops the picks, as does a new frame.

Save and apply… offers the frame's name with `.poni`. The frames after those already queued are integrated with the new PONI; the Image tab's rings, the Integration box and an Integrator's start follow it when it takes effect.

SPEC console, Commands, Messages



The SPEC console

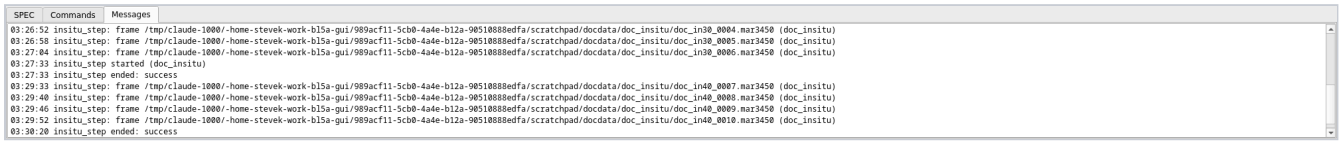
SPEC shows what SPEC prints, as spec-ioc's `Log` PV carries it, the plans' output included. `--- N lines not shown ---` marks a burst longer than the PV holds between two updates; `--- spec-ioc restarted ---` marks an IOC restart.

The entry sends one SPEC command (Enter or **Send**) when the queue is idle (see Who drives SPEC). The command is echoed as `HH:MM:SS SPEC> command`, followed by SPEC's reply. A command that does not end OK adds a line with its status; one that could not be written to SPEC, a line with why instead of a reply. **^C** interrupts the command sent from here; it never touches a plan's command.



The commands the plans sent

Commands lists every SPEC command a plan sent, with the time and the plan, and its reply indented below.



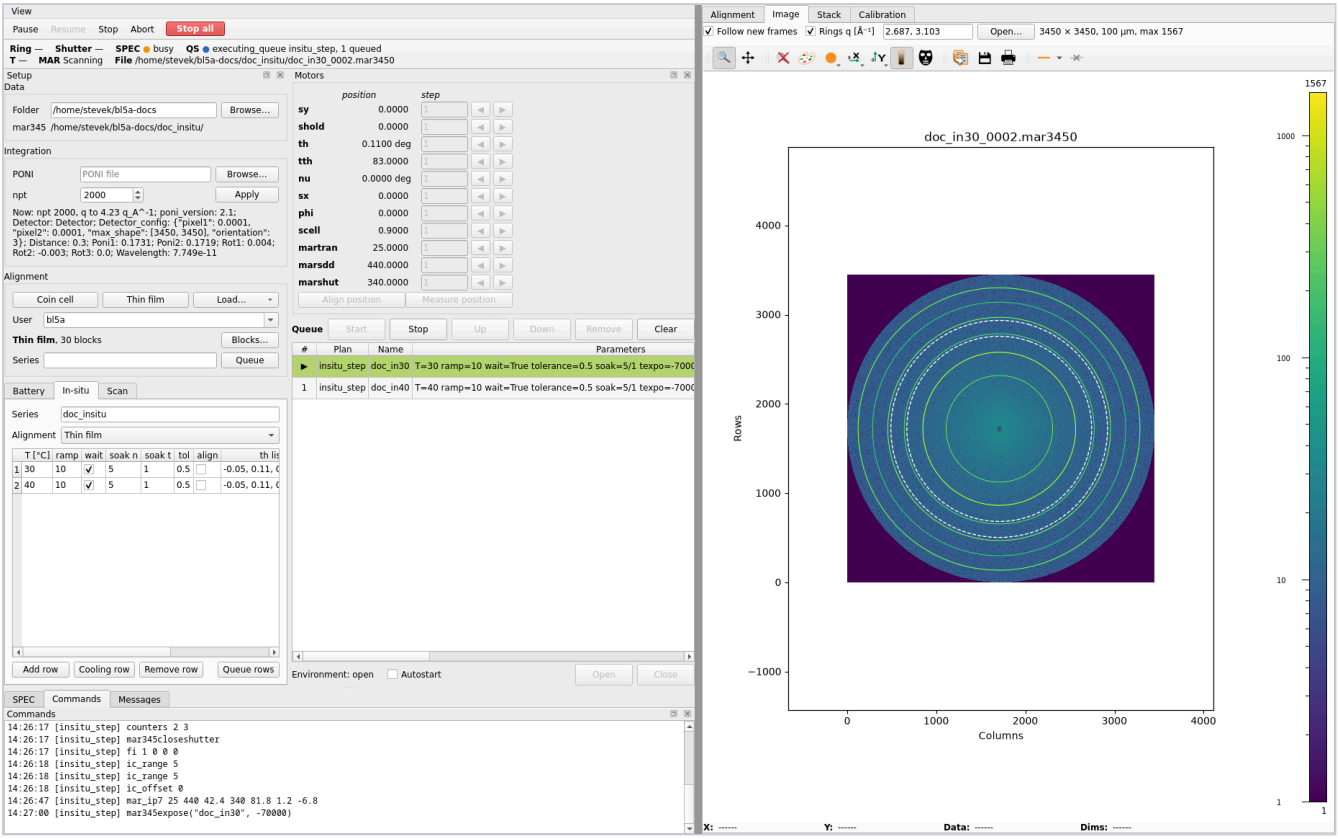
The GUI's messages

Messages is the GUI's own log: each queue request and its result, each plan's start, each queue item's end with its exit status (and, when it did not succeed, why), each frame with its series, and errors from bl5a-qs and the integrator. A request bl5a-qs refuses also opens a "bl5a-qs" box.

Walkthrough: a battery cycle

1. Mount the coin cell. Check the **Data** folder. In **Alignment**, click **Coin cell** (or **Load…** a sequence of your own); its blocks open in the Blocks window, to change or just close. Type a series (for example the cell's name) and click **Queue**.
2. Click **Start** in the queue. The Alignment tab draws the sy and shold fly scans; the history gets a row with the sy and shold the run left.
3. With the queue idle, type `umv tth 83` in the SPEC console.
4. On the **Battery** form: a file name, Texpo, **Repeat** with N and Ti (or **Single shot**), the geometry and the same series, so the frames go to the alignment's folder. **Add to queue**, answer **Yes** to the beamstop question, then **Start**.
5. Start the cycler from its own computer.
6. Each frame shows on the **Image** tab and is added to the **Stack** tab; choose `time [min]` for y to follow the cycle.
7. To end before N frames, click **Stop after frame**.

Walkthrough: a film against temperature



An in-situ run: the frame, the motors, the queue and the commands

1. Mount the film. Check the **Data** folder and, for the detector position `mymar_in` uses, the **Integration** PONI. On the **In-situ** form, type the series and pick the Alignment sequence.
2. Fill one row per temperature: T, ramp, wait and tol, a soak if wanted, align, the th list, texpo and the file name. Add the **Cooling row** last.
3. **Queue rows**, then **Start**.
4. Each row sets the temperature, aligns (Alignment tab) and takes its frames (Image and Stack tabs). The history gets a row per aligned step with its T.
5. On the **Stack** tab, pick one th (for example `th 0.13`) to compare the same angle across temperatures.

AI control (MCP)

`gui/run_mcp.py` gives Claude Code or Codex the GUI's controls as MCP tools (setup: install.md). The client starts it through `deploy/bl5a-mcp` when it starts in the `bl5a_gui` directory. It builds queue items with the forms' own code, passes the same gate as the buttons and writes only the PVs the GUI writes, so an item the AI queues is the item the form would queue. It runs beside the GUI, and the GUI shows what the AI queued and runs.

Ask in plain words, for example "queue a thin-film alignment for series film3 and start it", "show me the last frame" or "what did the last fly scan find?".

Tools	What they do	In the GUI
`status`	the queue's and SPEC's state, and `allowed`: the actions open now	status line; which buttons are live

Tools	What they do	In the GUI
`queue`, `history`	the waiting items; the finished ones with their exit status	Queue
`motors`	position, step and unit of each motor	Motors
`scans`, `scan`, `alignment`	the scans with their peak values and where each moved its motor; the stages and the alignment history	Alignment tab
`messages`, `spec_log`, `commands`	the three logs	Messages, SPEC, Commands
`frame_image`	the last frame, or a named one, as a picture	Image tab
`integration`, `stacks`, `stack`	the integrator's PONI and npt; the stacks; one stack's rows	Integration, Stack tab
`alignment_sequences`, `alignment_sequence`, `save_alignment_sequence`	a user's sequences; one sequence's blocks; save one as the user's	Alignment box: Load…; Blocks window: Save as...
`queue_alignment`, `queue_scan`, `queue_mar_series`, `queue_insitu`	queue items	Alignment Queue, Scan form, Battery form, In-situ form
`queue_remove`, `queue_move`, `queue_clear`, `queue_start`, `queue_stop`, `queue_autostart`, `environment_open`, `environment_close`	edit and run the queue	the queue's buttons
`pause`, `resume`, `stop`, `abort`, `stop_after_frame`	end or hold the running plan	Run buttons, Stop after frame
`tweak`	move a motor by a step	Motors ◀ ▶
`spec_command`, `spec_result`, `spec_abort`	one SPEC command, its result, ^C	SPEC console
`stop_all`	SPEC's ^C for every motor, count and command, and an abort of the plan	Stop all
`integration_set`	a PONI file and npt for the next frames	Integration Apply

The rules are the GUI's:

- An action that is not open now is refused with the reason, where the button would be grey: `pause: not open now (qs idle; …)`. The tools ask bl5a-qs first, so each one sees what the previous one did.
- `data_folder` left out uses the GUI's Data folder; `""` uses bl5a-qs's `--data-root`; an absolute path is used as it is. The data then goes to `<folder>/<series>`.
- `queue_mar_series` needs `beamstop_in_place: true`, the GUI's beamstop question. The server tells the AI to ask you first, and to ask before `abort`, `stop_all` and `queue_clear`. Your MCP client's permission prompt is the other guard: allow only the read tools without a prompt.
- `pause` then `resume` fails the plan, as in the GUI. `stop_after_frame` ends a series after the frame being taken; asked before the first frame starts, the series stops with none.

- `spec_command` waits up to `wait_s` (default 30 s); a longer command comes back as running, and `spec_result` gives its end. A command whose write to SPEC fails is an error, or, once it has come back as running, an `error` in place of `status` and `reply` in `spec_result`; the MCP server's log has it too. What SPEC prints (the table of `wm`, for example) goes to the SPEC log, not the reply: read it with `spec_log` from the `spec_log_since` the result gives.
- The logs hold what happened since the server started. Each read gives `last`; pass it as `since` next time to get only the new lines.

As on the Battery form, move tth to 83° or more before a standard `queue_mar_series` (Battery measurements); a thin-film alignment leaves it near 1.2. Use `tweak`, or `spec_command` with `umv tth 83` at the beamline.

When something is wrong

What you see	What it means	What to do
QS ● not answering	bl5a-qs does not reply on `--qs-control`	`systemctl --user status bl5a-qs`, its journal (`journalctl --user -u bl5a-qs`), or the address
SPEC —	spec-ioc's PVs do not connect	`systemctl --user status bl5a-spec-ioc`; check Channel Access to its host
SPEC ● disconnected	spec-ioc runs but SPEC's server does not answer it	start SPEC with `-S`; check the host and port in `SPEC=`
Motor buttons and the console stay grey	a plan runs, SPEC is busy, or a console command runs	wait, or stop the queue
"Battery", "In-situ" or "Alignment" box	a field breaks a rule, a series that is not a folder name included	the box names the field and the rule
"bl5a-qs" box	bl5a-qs refused a request	the box gives its reason
Messages: "... ended: fail: ..." and Autostart unticked	the item failed; the text after `fail:` says why	the last SPEC command and its reply are at the bottom of Commands
Image info line shows a file error	the frame's path is not readable here	mount the mar345 data directory at the same path on this computer
Stack: "integrator: not answering"	bl5a-integrator is not running or not at `--integrator-rep`	`systemctl --user status bl5a-integrator`; Refresh
"Data" box	the Data folder is not an absolute path, holds a space, or with the series is over 200 bytes	the box gives the rule
Data: mar345 folder "missing"	the mar345 IOC does not see the item's folder	mount the data root at the same path on the IOC's host
Messages: "... ended: fail: device state error: data folder: ..."	the data root does not exist on bl5a-qs's host	create it, or fix the Data folder and queue again
Integration: "Refused: ..."	the integrator could not use the PONI or npt	the text after "Refused:" says why
T reads "—"	no temperature PVs are configured	the temperature is set by SPEC's macros; read it in SPEC